CHAPTER 10

HOSTING

10.1 Hosting Overview

The **Personalized Learning Content Generator (PLCG)** system is deployed as a **web-based application** using a combination of:

- **FastAPI** backend (model/API layer)
- **Streamlit or HTML/JS frontend**
- **Cloud hosting platforms such as Render/Streamlit Cloud/HuggingFace Spaces**

This deployment enables users to generate explanations, examples, MCQs, flashcards, and personalized study material **directly from their web browser**, without installing any software.

10.2 Deployment Steps

Step 1: Prepare the Project Directory Structure

```
plcg/
│── backend/
│    ├── app.py
│    ├── sample_content.json
│    ├── requirements.txt
│    └── utils/
│
│── frontend/
│    ├── app.py
│    ├── assets/
│
│── models/
│── data/
```

Step 2: Upload the Project to GitHub

- Push the backend, frontend, models (if small), and UI files to a GitHub repository.
- Include a proper requirements.txt or environment.yml.

Step 3: Choose a Hosting Platform

You can deploy using:

Option A — Render.com (Recommended for API + UI)

Easily hosts FastAPI + Streamlit apps.

Option B — Streamlit Cloud

Ideal for hosting only the **frontend**, while backend runs on another platform.

Option C — HuggingFace Spaces

Best for hosting complete ML demos with CPU/GPU support.

Step 4: Deploy Backend (FastAPI)

Example steps on Render:

1. Create a new **Web Service**.
2. Connect GitHub repository.
3. Choose build command:
4. pip install -r backend/requirements.txt
5. Choose run command:
6. uvicorn backend.app:app --host 0.0.0.0 --port 8000
7. Deploy the service.

This provides a public API endpoint for the frontend.

Step 5: Deploy Frontend (Streamlit or HTML/JS)

For Streamlit Cloud:

1. Sign in to https://share.streamlit.io
2. Click **Deploy App**
3. Select the GitHub repo and Streamlit entry file (frontend/app.py)
4. Deploy

Streamlit automatically builds and hosts the application.

Step 6: Link Frontend to Backend

- Update frontend to point to the hosted FastAPI URL:
- https://your-backend-service.onrender.com/generate_gpt5
- Test end-to-end generation.

10.3 Technology Used for Hosting

- **Render / Streamlit Cloud / HuggingFace Spaces**
  Hosting platforms for backend and frontend services
- **GitHub**
  Source code repository and version control
- **FastAPI + Uvicorn**
  Backend API server
- **Streamlit / HTML-JS**
  User interface
- **Python 3.10+**
  Runtime environment
- **pip / virtualenv**
  Package and environment management

10.4 Advantages of Cloud Hosting

- **Zero installation** required for users
- **Cross-platform access** from mobile and desktop browsers
- **Automatic scalability** based on traffic load
- **HTTPS support** for secure communication
- **Continuous deployment** — GitHub updates automatically reflect in hosted app
- **Modular hosting** (frontend and backend can be hosted separately)

10.5 Alternative Hosting Options

- **Heroku** (simple deployment for small apps)
- **Google Cloud Run** (serverless API hosting)
- **AWS EC2 / Lightsail** (full control over server environment)
- **Azure App Service**
- **Dockerized self-hosting** on Linux or VPS

10.6 Hosting Requirements

Minimum Requirements

- 2–4 GB RAM server instance
- Python 3.9–3.11 support
- Stable internet connection
- HTTPS-enabled hosting platform